

Subject: Manual, Two step FVE method

From : Jan Mooiman

Date : 2026-08-03 21:33:10

1 Description of 1D input file

Five observation points are hard coded in the software, these are located at:

- 1 the centre of the grid.
- 2 at the west boundary.
- 3 at the east boundary.
- 4 halfway to the west boundary.
- 5 halfway to the east boundary.

Extra observation points can be specified in the input file according the description in [section 5](#)

2 File description Burgers & Advection-diffusion

Items in the color red are only to be used in the development stage of the software. An example input file is given in [section 6](#).

Table 1: Standard input-file with default settings.

Keyword	Default	Description
logging	"None"	"iterations" "matrix" "pattern"
[Boundary]		
treg	150.0	Regularization time boundary signal
eps_bc_corr	0.01	Boundary correction term to force the boundary value to the given boundary value
bc_signals	[· , ·]	["constant" "sine"]
bc_type	[· , ·]	["dirichlet" "borsboom"]
bc_vals	[· , ·]	[Real value]
[Domain]		
Lx	-	Length of the domain
x_origin	-	x location of the origin
[Initial]		
ini_vals	[· [, ·]]	[real value, real value] If one value is given it is assumed constant over the whole domain
ini_vars	"constant"	["constant", "linear"]

(continued on next page)

(continued from previous page)

Keyword	Default	Description
[Numerics]		
dt	5.0	Time step size [s], if dt == 0: then stationary simulation
theta	0.5	Implicitness factor ($0.5 \leq \theta \leq 1.0$)
c_psi	4.0	Smoothness factor used for regularization
iter_max	25	Maximum number of nonlinear iterations
eps_newton	1e-12	Accuracy of the Newton solver (non-linear)
eps_bicgstab	1e-12	Accuracy of the linear solver
eps_abs_function	0.01	Smoothing factor for the absolute-function
linear_solver	"bicgstab"	Type of linear solver "bicgstab", "multigrid"
regularization_init	false	true, false
regularization_iter	false	true, false
regularization_time	false	true, false
[Output]		
dt_his	dt	Time interval to write the history file, [s]
dt_map	60.0	Time interval to write the map file, [s]
[Physics]		
do_convection	false	true, false
do_viscosity	false	true, false
viscosity	0.01	Diffusion/Viscosity coefficient
do_source	false	true, false
src_type	"none"	"none", "quadratic"
[Time]		
tunit	"s"	Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d"
start	0.0	Start time of the simulation, [tunit]
stop	60.0	Stop time of the simulation, [tunit]
[[ObservationPoint]]		
x	-	x -coordinate of the observation point
y	-	y -coordinate of the observation point
name	"..."	Name of observation point between double quotes

3 Example input 1D file Advection-diffusion

```
Logging = "None" # "iterations", "matrix", "pattern"
```

```
[Boundary] # west, east
treg       = 1800.0 # Regularization time boundary signal [s]
eps_bc_corr = 0.01
bc_signals  = ["constant", "constant"]
bc_type     = ["dirichlet", "borsboom"]
bc_vals     = [2.0, 0.0]
```

```
[Domain]
```

Manual, Two step FVE method

```
Lx          = 20000.0 # Domain length [m]
x_origin    = 0.0    # Origin of the x-coordinate [m]

[Initial]
ini_vals    = [1.0]
ini_vars    = ["constant"]

[Numerics]
dt           = 1.0    # Time step size [s], if dt == 0: then stationary problem
dx           = 1.0    # Grid size [m]
c_psi       = 4.0
theta       = 0.50    # Implicitness factor (0.5 <= theta <= 1.0)
iter_max     = 25.0    # Maximum number of non-linear iterations
eps_newton   = 1e-12
eps_bicgstab = 1e-12
eps_abs_function = 0.01
regularization_init = true
regularization_iter = false
regularization_time = true

[Output]
dt_his      = 1.0
dt_map      = 300.0
dt_screen   = 300.0

[Physics]
do_convection = true
do_viscosity  = true
viscosity     = 0.1
do_source     = false
src_type      = "none"

[Time]
tunit        = "s"    # "s", "m", "h", "d"
tstart       = 0.0
tstop        = 14400.0

# Hard coded observation points are not listed

[[ObservationPoint]]
x            = 2500
y            = 0
name         = "_a"

[[ObservationPoint]]
x            = 7500
y            = 0
name         = "_b"

[[ObservationPoint]]
x            = 12500
y            = 0
name         = "_c"

[[ObservationPoint]]
x            = 17500
y            = 0
name         = "_d"
```

4 Description of 2D input file

Nine observation points are hard coded in the software, these are located at:

- 1 the centre of the grid.
- 2 in the middle of the north boundary.
- 3 in the middle of the east boundary.
- 4 in the middle of the south boundary.
- 5 in the middle of the west boundary.
- 6 in the North-East corner of the grid.
- 7 in the South-East corner of the grid.
- 8 in the South-West corner of the grid.
- 9 in the North-West corner of the grid.

Extra observation points can be specified in the input file according the description in [section 5](#).

5 File description

Items in the color red are only to be used in the development stage of the software. An example input file is given in [section 6](#).

Table 2: Standard input-file with default settings.

Keyword	Default	Description
logging	"None"	"iterations" "matrix" "pattern"
[Boundary]		
treg	150.0	Regularization time boundary signal
eps_bc_corr	0.01	Boundary correction term to force the boundary value to the given boundary value
bc_type	[· , · , · , ·]	["free-slip" "no-slip" "borsboom" "mooiman"]
bc_vars	[· , · , · , ·]	["-" "zeta" "q"]
bc_vals	[· , · , · , ·]	[Real value]
[Domain]		
bed-level-file	-	Name of bed-level-file between double quotes
mesh-file	-	Name of mesh file point between double quotes

(continued on next page)

(continued from previous page)

Keyword	Default	Description
[Initial]		
ini_vars	None	["zeta", "zeta_gauss_hump" "zeta_constant" "zeta_gauss_hump_x" "zeta_gauss_hump_y"]
gauss_amp	0.0	
gauss_mu	0.0	
gauss_mu_x	0.0	
gauss_mu_y	0.0	
gauss_sigma	350.0	
gauss_sigma_x	350.0	Optional if "gauss_sigma" is not specified
gauss_sigma_y	350.0	Optional if "gauss_sigma" is not specified
[Numerics]		
dt	5.0	Time step size [s], if dt == 0: then stationary simulation
theta	0.501	Implicitness factor (0.5 <= theta <= 1.0)
c_psi	4.0	Smoothness factor used for regularization
iter_max	25	Maximum number of nonlinear iterations
eps_newton	1e-12	Accuracy of the Newton solver (non-linear)
eps_bicgstab	1e-06	Accuracy of the linear solver
eps_abs_function	0.01	Smoothing factor for the absolute-function
linear_solver	"bicgstab"	Type of linear solver "bicgstab", "multigrid"
regularization_init	false	true, false
regularization_iter	false	true, false
regularization_time	false	true, false
[Physics]		
do_linear_waves	true	true, false
do_continuity	true	true, false
do_q_equation	true	true, false
do_r_equation	true	true, false
do_convection	false	true, false
do_bed_shear_stress	false	true, false
do_viscosity	false	true, false
chezy_coefficient	25.0	Bed friction coefficient
viscosity	0.01	Diffusivity coefficient
[Output]		
dt_his	dt	Time interval to write the history file, [s]
dt_map	60.0	Time interval to write the history file, [s]
[Time]		
tunit	"s"	Unit of the start and stop time. Values seconds "s", minutes "m", hours "h" and days "d"
start	0.0	Start time of the simulation, [s]
stop	60.0	Stop time of the simulation, [s]

(continued on next page)

(continued from previous page)

Keyword	Default	Description
[[ObservationPoint]]		
x	-	x -coordinate of the observation point
y	-	y -coordinate of the observation point
name	"..."	Name of observation point between double quotes

6 Example input 2D file

```

logging = "None" # "iterations", "matrix", "pattern"

[Boundary] # north, east, south, west
treg      = 150.0 # Regularization time boundary signal
eps_bc_corr = 0.01
bc_type    = ["borsboom", "borsboom", "borsboom", "borsboom"] # Type "neumann", "dirichlet", "borsboom"
bc_vars    = ["zeta", "zeta", "zeta", "zeta"]
bc_absorbing = [true, true, true, true]
bc_vals    = [0.0, 0.0, 0.0, 0.0]

[Domain]
mesh_file      = "geometry/62x62_6x6km_net.nc"
bed_level_file = "geometry/62x62_6x6km_flat.dep"

[Initial]
ini_vars      = ["zeta", "zeta_gauss_hump"]
Gauss_amp     = 1.0
Gauss_mu      = 0.0
Gauss_mu_x    = 0.0
Gauss_mu_y    = 0.0
Gauss_sigma   = 350.0
Gauss_sigma_x = 350.0
Gauss_sigma_y = 350.0

[Numerics]
dt              = 5.0 # Time step size [s], if dt == 0: then stationary problem
theta           = 0.501 # Implicitness factor (0.5 <= theta <= 1.0)
c_psi           = 4.0
iter_max        = 25 # Maximum number of nonlinear iterations
eps_newton      = 1e-12
eps_bicgstab    = 1e-06
eps_abs_function = 0.01
linear_solver    = "bicgstab" # "bicgstab", "multigrid"
regularization_init = false
regularization_iter = false
regularization_time = false

[Physics]
do_linear_waves = true
do_continuity   = true
do_q_equation   = true
do_r_equation   = true
do_convection   = false
do_bed_shear_stress = false
do_viscosity    = false
chezy_coefficient = 25.0
viscosity       = 100.0

[Output]
dt_his = 0.01
dt_map = 30.0

[Time]
tunit = "s"
tstart = 0.0
tstop = 1800.0

```

